

Scope of Work

This project aims to develop an automated software tool for the Nebraska Department of Transportation (NDOT) to support the implementation of the Balanced Mix Design (BMD) framework for asphalt mixture approval. The current NDOT workflow relies heavily on manually transferring raw laboratory data into large Excel spreadsheets to perform engineering calculations and generate performance evaluation metrics. This process is time-consuming, difficult to scale, and vulnerable to human error. The purpose of this project is to convert the existing Excel-based workflow into a modular Python-based application capable of automatically processing raw testing data and generating engineering outputs with minimal user intervention.

The project will utilize datasets obtained from the provided AutoBMD.xlsx workbook and associated laboratory test outputs. These datasets include aggregate gradation measurements, volumetric properties, IDEAL-CT cracking test data, and Hamburg Wheel Tracking Test (HWTT) rutting data. Additional reference material will include NDOT specification limits, control point requirements, and engineering equations currently embedded within the Excel workbook. Python libraries such as pandas, NumPy, matplotlib, and SciPy will be used to organize data, perform calculations, and generate visualizations required for engineering analysis.

The software will be divided into four independent but connected modules. The first module, the Aggregate Gradation Automator, will read aggregate sieve percentage data and user-defined blend ratios to calculate a combined gradation curve. The program will compare the resulting gradation against NDOT control point specifications and generate a sieve size versus percent passing plot to determine whether the blended aggregate falls within the acceptable specification range.

The second module, the Volumetric Properties Calculator, will process measurements from the volumetrics section of the dataset and implement the equations currently used in the Excel workbook to calculate required volumetric properties. The module will automatically evaluate quality control requirements, including verification that Air Voids remain within the acceptable specification range of 6.5% to 7.5%. If the calculated values fall outside of this range, the software will flag the sample as invalid for performance testing. The output will include a summary table containing calculated properties, validation results, and warning messages when applicable.

The third module, the IDEAL-CT Analyzer, will process raw time-series load versus displacement data collected during cracking tests. The software will automatically calculate the CTIndex by determining the area under the load-displacement curve,

identifying the displacement at 75% of the peak load, and calculating the post-peak slope. The module will generate clean engineering plots of load versus displacement and provide the calculated CTIndex values for each tested sample.

The fourth module, the Rutting Analyzer, will evaluate Hamburg Wheel Tracking Test (HWTT) data using wheel cycles and rut depth measurements. The software will implement an algorithm to identify the Stripping Inflection Point (SIP), defined as the location where the rutting slope changes significantly due to moisture damage. The module will also determine the exact cycle at which the rut depth exceeds the 12.5 mm failure threshold and will evaluate whether the asphalt mixture survives the required 20,000 loading cycles. Outputs will include rut depth versus cycle plots with clearly identified SIP and failure points, along with a final pass/fail determination for each sample.

The completed software will be designed as a fully functional Python script or Jupyter Notebook capable of executing all four modules sequentially. Emphasis will be placed on modular code structure, automation, reproducibility, and ease of use for NDOT engineers. In addition to the source code, a short user guide will be created describing the required input formatting, expected data organization, and instructions for operating the software tool.

Finally, the completed application will be evaluated based on its ability to accurately reproduce the engineering calculations currently performed in Excel while improving efficiency, consistency, and scalability. The final product is intended to provide NDOT engineers with a repeatable and reliable workflow for asphalt mixture performance analysis under the Balanced Mix Design framework.